

South Asia's Climate Emergency

Floods, Heatwaves, and the Subcontinent's Civilisational Existential Risk

Blue Lotus Research Intelligence | August 2026

Classification: Open Intelligence

Executive Summary

South Asia is the world's most climate-vulnerable region. Home to 1.9 billion people — nearly 25% of humanity — on approximately 5% of the world's land surface, the subcontinent faces a convergence of climate threats that are simultaneously more severe, more rapid, and more socially devastating than anywhere else on earth. The IPCC Sixth Assessment Report (2021-2022) identifies South Asia as one of the regions facing the highest physical climate risks, with the Hindu Kush Himalayan water tower melting, the Bay of Bengal intensifying tropical cyclones, the monsoon system becoming more erratic and extreme, and wet-bulb temperatures in parts of Pakistan, India, and Bangladesh approaching or exceeding human physiological survival limits. The World Bank's 2021 "Groundswell" report projects that South Asia could generate 216 million internal climate migrants by 2050 under pessimistic scenarios — more than any other world region — creating displacement pressures that will reshape the subcontinent's political geography within a single generation.

The climate emergency is not a future projection for South Asia — it is already happening. Pakistan's catastrophic 2022 monsoon floods submerged one-third of the country's land area, affected 33 million people, killed 1,739 (official count), destroyed 2 million homes, and caused \$30 billion in damage — a disaster whose probability was made 8 times more likely by climate change (World Weather Attribution study). India's heat wave of April-May 2022 killed thousands (precise toll underreported), with wet-bulb temperatures in Uttar Pradesh, Rajasthan, and Andhra Pradesh breaching 35°C — the physiological threshold beyond which the human body cannot cool itself through sweating regardless of shade or rest. Bangladesh experiences a major cyclone striking its coast every 2-3 years; its delta — home to 30 million people below 1 metre sea level — is among the most flood-exposed human settlements on earth. The climate emergency is not coming. It has arrived.

Part I — The Monsoon System: South Asia's Civilisational Lifeline Under Stress

The Monsoon as the Engine of South Asian Civilisation

The Indian summer monsoon — which delivers 70-90% of annual precipitation to the Indian subcontinent between June and September — is not merely a weather pattern. It is the hydraulic foundation of South Asian civilisation: the water source for 800+ million smallholder farmers, the recharge mechanism for the Indo-Gangetic aquifer system, the timing signal for the agricultural calendar from Punjab to Bengal, and the cultural and spiritual anchor of South Asian life across 5,000 years of human settlement. When the monsoon fails, civilisations collapse — as documented in the Harappan collapse (~1900 BCE), the

mega-droughts of the Medieval Warm Period, and the colonial-era famines of 1876-1879 and 1899-1900 that killed 12-29 million Indians.

The monsoon's mechanics are driven by differential heating between the land (South Asian subcontinent) and the sea (Indian Ocean), creating a pressure gradient that draws moist air northward. Climate change is disrupting this system in complex, non-linear ways: the land-sea temperature differential is changing as the land warms faster than the ocean; the warming Arabian Sea is intensifying and prolonging pre-monsoon cyclone season; the weakening land-sea gradient is making monsoon onset dates less predictable; and the increase in atmospheric moisture content (approximately 7% more water vapour per degree of warming, per Clausius-Clapeyron thermodynamic relationship) is making rainfall events more intense even as their spatial distribution becomes more erratic. The result: "feast or famine" monsoons — more extreme floods and more intense droughts, sometimes in the same season and the same country.

The Monsoon Record: Increasing Extremes

India Meteorological Department (IMD) data from 1901-2025 shows the following trend in Indian Summer Monsoon Rainfall (ISMR):

- No statistically significant trend in total seasonal rainfall nationally — the monsoon is not "getting drier" on average
- Significant increase in the intensity of extreme rainfall events (daily rainfall exceeding 100mm, 200mm, 300mm thresholds)
- Significant decrease in moderate rainfall days (10-50mm/day) — the consistent "soaking" rains that recharge groundwater and support sustained agricultural growth
- Increasing spatial variability — monsoon patterns are becoming more concentrated, leaving some regions in drought while others flood

The practical agricultural consequence: crops that depend on consistent moderate rainfall during critical growth stages (transplantation, grain filling) are being disrupted by the shift from moderate to extreme patterns. CIMMYT (International Maize and Wheat Improvement Center) estimates that Indian wheat yields will decline 6-25% by 2050 even under moderate warming scenarios as growing season temperatures exceed critical thresholds during flowering. Rice yields in Bangladesh are projected to decline 8-17% by 2050 from combined flooding and drought stress.

Part II — Extreme Heat: When the Planet Becomes Uninhabitable

Wet-Bulb Temperature: The Physiological Limit

The critical climate threat that distinguishes South Asia from other vulnerable regions is not merely heat, but wet-bulb temperature (TW) — the combined measure of temperature and humidity that determines whether the human body can cool itself through sweating. When TW exceeds approximately 32-34°C, even healthy adults at rest in shade begin to accumulate body heat; at TW > 35°C (equivalent to 40°C at 100% humidity, or approximately 50°C at lower humidity), core body temperature rises uncontrollably within 6 hours regardless of shade, water, or air circulation, causing organ failure and death.

The Raymond et al. (2020) study, using global weather station data since 1979, found that locations reaching TW > 35°C — previously considered a theoretical future threshold — had already been exceeded in multiple South Asian locations. The Indus River valley (Pakistan, India's Rajasthan border), the Persian Gulf coast, and coastal Bangladesh have all recorded brief TW > 35°C exceedances. As the IPCC AR6 projects, under high emission scenarios (SSP5-8.5), large portions of South Asia's most densely populated agricultural regions — the Ganges-Brahmaputra-Meghna delta (Bangladesh, West Bengal), the Indus valley (Pakistan, Punjab India), and coastal lowlands — will experience TW > 35°C conditions for sustained periods (days to weeks) by the 2050s-2060s, potentially rendering outdoor

agricultural labour — the livelihood of 400-600 million South Asians — physiologically impossible during peak growing seasons.

India's Heat Wave Crisis

India has experienced severe heat waves in every year since 2015. The April-June 2022 heat wave was exceptional:

- New Delhi recorded 49°C on May 15, 2022 — the highest temperature recorded in the city in recorded history
- Wet-bulb temperatures in parts of Uttar Pradesh, Madhya Pradesh, and Rajasthan reached 35°C
- The official death count was 90 people, but peer-reviewed excess mortality analysis (The Lancet, 2023) estimates actual excess deaths in the range of 3,500-4,000 in the April-June 2022 period in India alone
- Power demand surged to record levels (207 GW peak demand April 29, 2022), triggering load shedding of 8-12 hours daily across major states, including the worst-affected regions — eliminating the cooling that air conditioning could otherwise provide
- Agricultural losses: wheat crop damaged in Rajasthan, Punjab, and Haryana as flowering-stage temperatures exceeded 35°C for 3-5 consecutive days in late March-April, potentially reducing yields by 15-25%

Pakistan's 2022 heat wave (concurrent with India's, April-May 2022) was even more extreme: Jacobabad, Pakistan, recorded 49°C with simultaneous high humidity — producing estimated TW values of 33-34°C — on multiple days in May 2022. Jacobabad has been described by climate scientists as among the "most frequently uninhabitable places on earth" during heat wave seasons — a designation that will, under current emission trajectories, extend to larger and larger portions of Pakistan's Punjab and Sindh provinces.

Part III — Flooding: When the Monsoon Becomes a Catastrophe

Pakistan 2022: The Flood That Submerged a Nation

The 2022 Pakistan monsoon floods represent the most catastrophic single climate event in South Asian recorded history. Statistics:

Scale: Approximately one-third of Pakistan's total land area (about 165,000 km²) was submerged at peak inundation — an area larger than the United Kingdom. The Indus River, already running above historical averages from Himalayan glacial melt, received extraordinary monsoon rainfall (200-400% above normal in Khyber Pakhtunkhwa, Balochistan, and Sindh provinces).

Human cost: 1,739 deaths (official count, widely acknowledged as significant undercount given the remoteness of affected areas); 33 million people displaced or affected (approximately 15% of Pakistan's population); approximately 7.9 million people internally displaced to camps or informal settlements; 2.1 million homes destroyed or severely damaged.

Economic cost: \$30 billion in damage and economic losses — approximately 10% of Pakistan's entire GDP in a single monsoon season. \$16 billion in direct asset losses (infrastructure, housing, crops). For a country already in IMF programme negotiations at the time, the flood was a fiscal catastrophe that required emergency international humanitarian assistance and directly contributed to the deepening economic crisis of 2022-2023.

Agricultural collapse: 2.2 million hectares of cropland submerged; cotton, rice, and wheat crops devastated; livestock losses of 800,000+ animals. The agricultural devastation drove food inflation to 40%+ in 2022-2023, directly worsening Pakistan's food security crisis.

World Weather Attribution's rapid attribution analysis found that climate change made rainfall of this intensity approximately 8 times more likely than it would have been in a pre-industrial climate — meaning this event, while natural in origin, has been made dramatically more probable by anthropogenic warming.

Bangladesh: The Delta at the Edge

Bangladesh occupies the Ganges-Brahmaputra-Meghna (GBM) delta — the world's largest river delta, where three of South Asia's great river systems converge before emptying into the Bay of Bengal. Approximately 80% of Bangladesh's 148,460 km² is floodplain; approximately 30 million Bangladeshis live within 1 metre of sea level; and approximately 60% of the country's agricultural land is inundated annually during normal monsoon floods — a controlled inundation that replenishes soil nutrients and supports Bangladesh's remarkable agricultural productivity. Climate change is turning this controlled inundation into an existential threat through three simultaneous mechanisms:

Sea level rise: The Bay of Bengal is rising approximately 3-4mm per year (global average) with additional contributions from groundwater subsidence (the GBM delta is sinking 5-20mm annually from groundwater extraction and sediment compaction) — creating an effective relative sea level rise of 10-20mm per year in coastal Bangladesh. Under RCP4.5 scenarios, 17% of Bangladesh's current land area could be permanently submerged by 2100, displacing approximately 18 million people; under RCP8.5, this rises to 20%+ of land and 20+ million displaced.

Cyclone intensification: The Bay of Bengal is warming faster than the global ocean average, creating more energy available for tropical cyclone intensification. Post-monsoon (October-December) cyclones striking Bangladesh's coast are intensifying more rapidly — reaching peak intensity closer to landfall and maintaining intensity longer. Cyclone Amphan (May 2020) — a super cyclone that struck the Sundarban coast of Bangladesh and West Bengal — caused \$13.5 billion in damage (India + Bangladesh combined) and was the most powerful cyclone to strike the Bay of Bengal in 20 years.

River flooding: GBM upstream discharge, already amplified by Himalayan glacial melt, is increasing during peak monsoon. China's Motuo dam on the Brahmaputra upstream creates an additional risk: sudden water release (whether accidental or operational) could inundate Assam (India) and then Bangladesh's northern plains within hours.

Part IV — The Agricultural Crisis: Feeding South Asia in a Warming World

The Crop Yield Threat

South Asia's 1.9 billion people are fed largely by the region's own agricultural system — unlike much of the Middle East or sub-Saharan Africa, which depend heavily on food imports. The "Green Revolution" that transformed South Asian agriculture (1960s-1980s) through high-yielding varieties, irrigation, and synthetic fertiliser has produced a food system capable of feeding the region's massive population. Climate change is systematically dismantling the conditions under which that food system operates.

Rice: The crop that feeds more South Asians than any other is highly temperature-sensitive during the pollination stage. Rice yield declines by approximately 10% for every 1°C increase in night-time temperature above 27°C during the reproductive stage. As South Asian night temperatures increase (minimum temperatures are rising faster than maximum temperatures due to greenhouse gas warming), rice yields are declining across Bangladesh, the eastern Indian states, and the Mekong delta. The IRRI projects rice production in South Asia to decline 10-20% by 2050 under moderate warming scenarios.

Wheat: India produces approximately 115 million metric tonnes of wheat annually, making it the world's second-largest producer after China and a critical food security buffer. Wheat is catastrophically sensitive

to heat stress during the flowering and grain-filling stages (late March-April). The 2022 heat wave — which brought temperatures above 40°C for 5+ consecutive days during wheat flowering in Punjab and Haryana — reduced the harvest by an estimated 3-5 million metric tonnes from projected levels. A similar event recurring 3-4 times per decade (as climate models project) would permanently reduce India's wheat buffer, potentially transforming India from a net wheat exporter to a net importer.

Water scarcity: All South Asian agricultural systems depend on groundwater as a buffer against monsoon variability. The Indo-Gangetic Plain, which produces 40%+ of South Asia's food, is drawing down its aquifer at 10-17mm per year (NASA GRACE satellite data). When the aquifer is depleted — a process that could occur within 25-50 years in the most intensively farmed areas — the agricultural system loses its buffer and becomes entirely dependent on monsoon reliability. In a world of more extreme monsoon variability, that dependency is catastrophic.

Part V — Climate Displacement: The Coming Migration Crisis

Internal Migration: The 2050 Scenario

The World Bank's 2021 Groundswell report, using integrated demographic-economic-climate models, projects that South Asia will experience 216 million internal climate migrants by 2050 under the pessimistic (high emissions, low development) scenario — the largest internal climate migration in any world region. Even under the optimistic scenario (moderate emissions, high development), South Asia generates 40 million internal climate migrants by 2050. The primary drivers, by region:

Bangladesh: 19.9 million (pessimistic) internal climate migrants by 2050, primarily from coastal flooding (cyclone and sea level rise), river flooding, and salinity intrusion destroying freshwater and agricultural capacity. The Bangladeshi diaspora experience — Bangladeshis have been migrating to Dhaka from climate-exposed coastal districts for 30 years, producing the world's most densely populated megacity (estimated 22-23 million in greater Dhaka) — is a preview of the displacement amplification that climate acceleration will produce.

India: 45 million internal migrants by 2050 (pessimistic scenario), primarily from water-stressed agricultural areas of Madhya Pradesh, Rajasthan, Chhattisgarh, and Jharkhand; heat-stressed coastal fishing communities of Andhra Pradesh and Odisha; and flood-affected river delta communities of West Bengal and Assam.

Pakistan: 32 million internal migrants by 2050 (pessimistic), driven by Sindh and Balochistan water scarcity, Indus flooding, and Karachi coastal flooding compounded by urban heat island effects.

For governments managing migration, infrastructure, and urban growth, the scale is staggering: these 216 million people are not hypothetical future statistics — they are farmers currently making marginal livelihood decisions in areas already experiencing climate stress, and the migration process is already underway.

Cross-Border Dimensions: India-Bangladesh as the Test Case

The India-Bangladesh border — a 4,156km fence largely completed under the India-Pakistan Border Security Force fencing programme — is the most significant climate migration pressure point in the world. The existing informal cross-border migration from Bangladesh to India (primarily Assam, West Bengal, and Tripura states) is a political flashpoint: the Assam citizenship register (National Register of Citizens, updated 2019) was designed precisely to identify "illegal Bangladeshi migrants" — generating a list of 1.9 million people facing statelessness, with many facing detention in "Foreigners' Tribunals." As climate displacement from Bangladesh accelerates, this migration pathway will carry millions more — creating a potential humanitarian and political crisis of extraordinary scale at India's northeastern border.

Conclusion: The Civilisational Stakes

South Asia's climate emergency is the defining challenge of the 21st century for the region — and one of humanity's defining challenges more broadly. A region of 1.9 billion people, with 400-600 million dependent on climate-sensitive agriculture, 30-50 million living within 1 metre of sea level, and entire river systems governed by a glacial water tower that is melting faster than any climate model predicted — faces a convergence of risks that no region has faced simultaneously.

The structural responses are real: India's 500 GW renewable energy target (195 GW installed August 2026); Bangladesh's Delta Plan 2100 (a 100-year integrated water management plan); Pakistan's Climate-Resilient Agriculture Policy (in implementation post-2022 floods); Nepal's hydropower development (3,000+ MW) providing clean energy to the subcontinent; the Coalition for Disaster Resilient Infrastructure (CDRI, India-hosted, 40-country membership). The scale of response has not yet matched the scale of risk.

The most important variable is the one South Asian countries cannot control unilaterally: global emissions trajectories. Under 1.5°C scenarios (Paris Agreement target), South Asia faces severe but manageable climate impacts. Under 3-4°C scenarios (consistent with current national commitments globally), South Asia faces crop system failure, permanent coastal inundation, and wet-bulb temperatures that make large portions of the subcontinent physiologically uninhabitable for outdoor work during summer months — a civilisational emergency, not merely an environmental one. The choices made in Beijing, Washington, Brussels, and New Delhi over the next decade will determine which South Asia 1.9 billion people inhabit by 2100.

Blue Lotus Research Intelligence | South Asia Series | Climate Emergency | August 2026

All data from IPCC AR6, World Bank, ICIMOD, IMD, Pakistan NDMA, and cited sources.